

Theoretical Ecotoxicology

Session 9: Toxicity prediction

University Duisburg-Essen SS 2025

Ralf B. Schäfer

Ecotoxicology, Faculty of Biology, UDE & RC One Health Ruhr

Intended learning outcomes of today's lecture

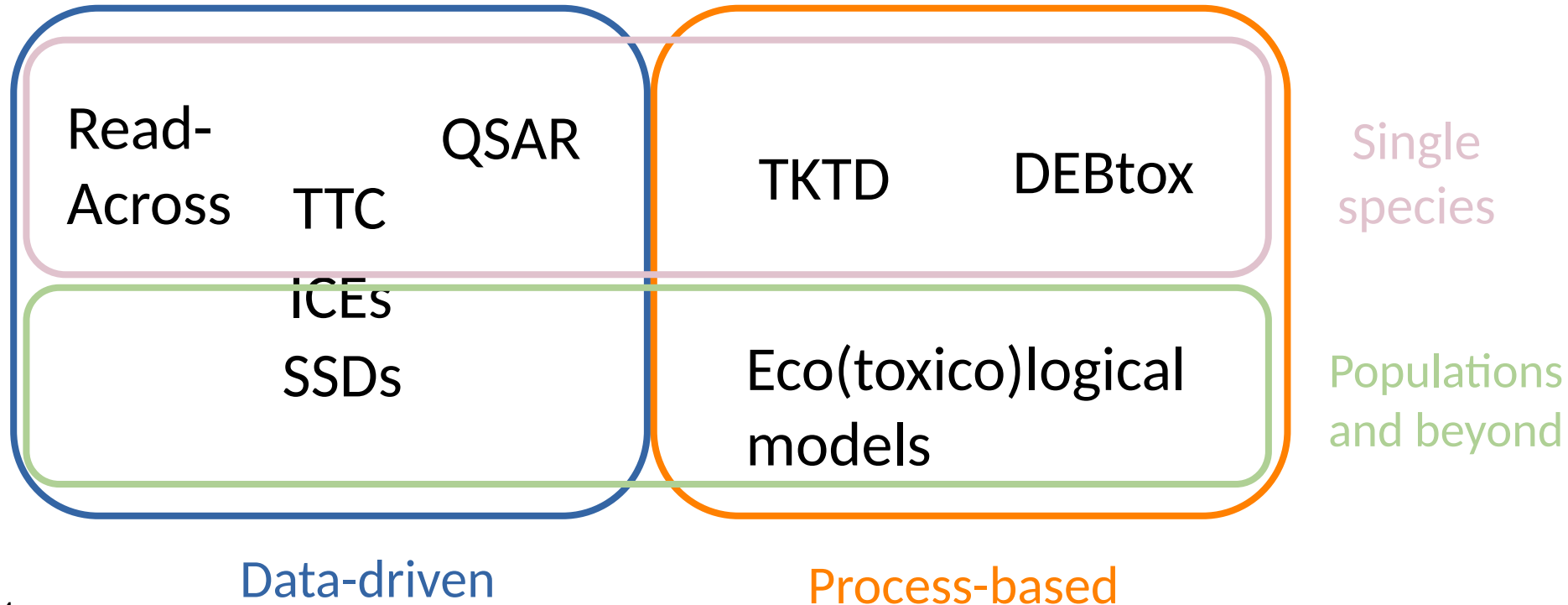
- Understanding the major concepts for toxicity prediction
- Knowledge on pre-processing steps and data sources
- Knowledge of tools for toxicity prediction

Learning targets and study questions

- Understanding the major concepts for toxicity prediction
 - Discuss the difference between QSAR and Read-Across.
 - Briefly describe what ICEs and ACRs are.
 - Which steps influence the comparability of toxicity predictions.
 - Discuss model validation and the role of splitting.
- Knowledge on pre-processing steps and data sources
 - Describe typical input formats for chemicals.
 - Provide an overview of data that can be useful to predict toxicity

In silico tools for ecotoxicity prediction

Particularly useful when empirical data is lacking



Quantitative Structure Activity Relationships (QSAR)

- Mathematical model
- General form: $\text{activity} = f(\text{structure})$
- Can be applied to estimate biological or physicochemical properties (e.g. persistence, bioaccumulation, distribution)
- May also consider biological predictors when predicting ecotoxicity

Quantitative Structure Activity Relationships (QSAR)

Example: Model for congeneric chemical

$$\log(1/C) = a(\log P) + b(\epsilon) + c(S) + d$$

C = effective concentration (e.g. EC50)

P = octanol water partitioning coefficient

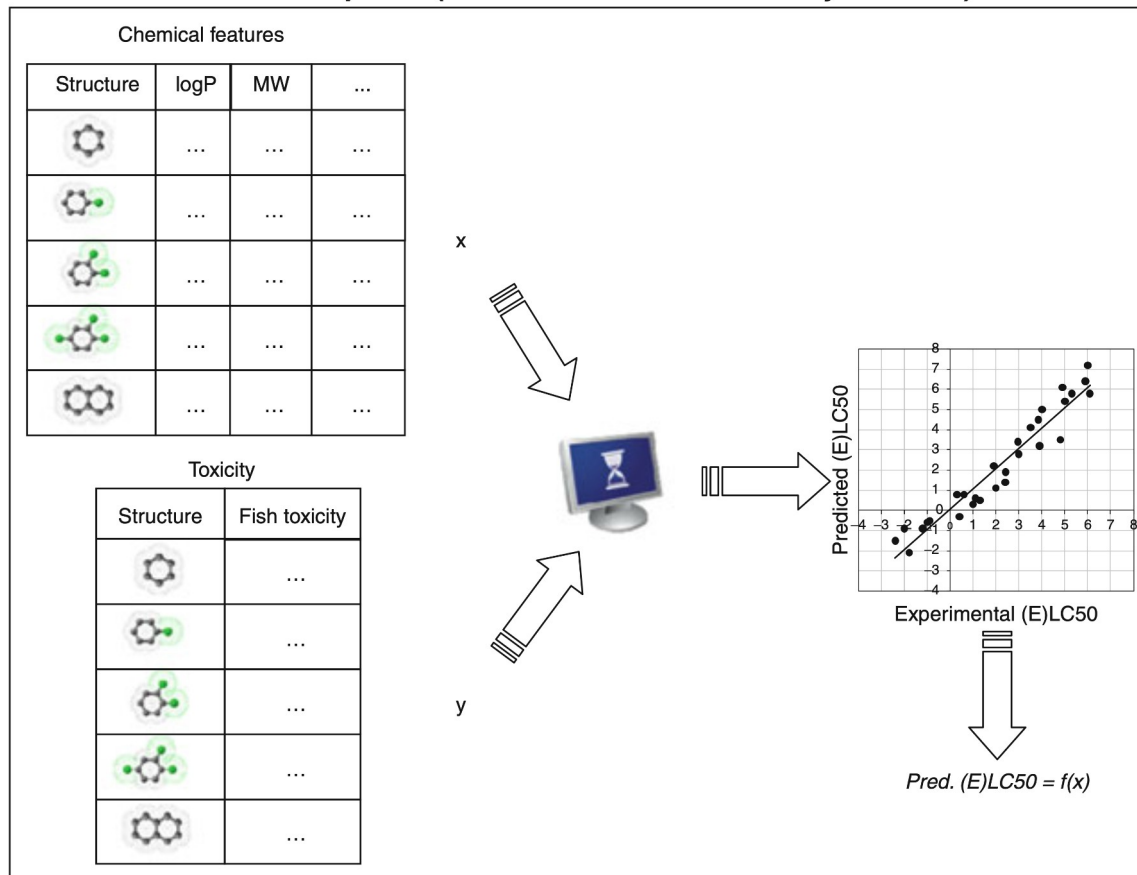
ϵ = Hammett substituent constant

S = Taft's substituent constant (steric)

a , b , c and d = model parameters/ functions for variables

General QSAR development

Model development (for substances whose toxicity is known).



General QSAR development

Model use (for substances with unknown toxicity).



QSAR
model

Prediction can be based on classical statistical (regression) or machine learning model

Input data for *in silico* approaches

Typical input formats (chemicals)

Identifier Type	Value
Chemical name	Caffeine
Molecular formula	C ₈ H ₁₀ N ₄ O ₂
SMILES	<chem>CN1C=NC2=C1C(=O)N(C(=O)N2C)C</chem>
InChI	InChI=1S/C8H10N4O2/c1-10-4-9-6-5(10)7(13)12(3)8(14)11(6)2/h4H,1-3H3
InChIKey	RYYVLZVUVIJVGH-UHFFFAOYSA-N
CAS number	58-08-2
PubChem CID	2519
IUPAC name	1,3,7-trimethylpurine-2,6-dione

<https://pubchem.ncbi.nlm.nih.gov/compound/2519>

Input data for *in silico* approaches

Typical input formats (chemicals)

For structural
representation

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For linking to databases

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Input data for *in silico* approaches

Typical chemical input data: Chemical descriptors

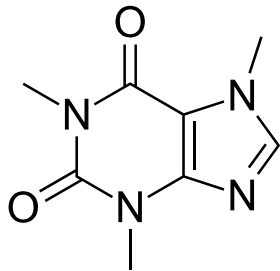
1-D

Scalar
physicochemical

- Molecular weight
- Log P
- Water solubility
- ...

2-D

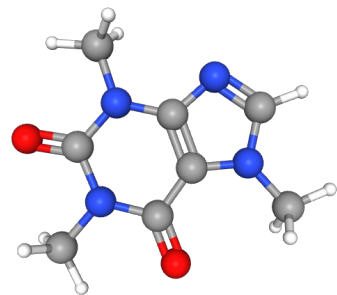
Topological/structural



- No. of atomic fragments (e.g. halogens, rings)
- Connectivity
- Polar surface area
- ...

3-D

Spatial/electron distribution



- Steric fields
- Electrostatic maps
- ...

Input data for *in silico* approaches

Typical chemical input data: Biological descriptors

Physiology

- volume-specific somatic maintenance
- maturation threshold for reproduction
- surface-area-specific max. assimilation
- ...

Phylogeny

e.g. **ape**, **taxize** R packages

Traits

- Feeding type
- Life span
- Reproductive rate
- ...

Biogeography

- Ecozone (e.g. geology, altitude)
- Climate
- ...

Add-my-Pet



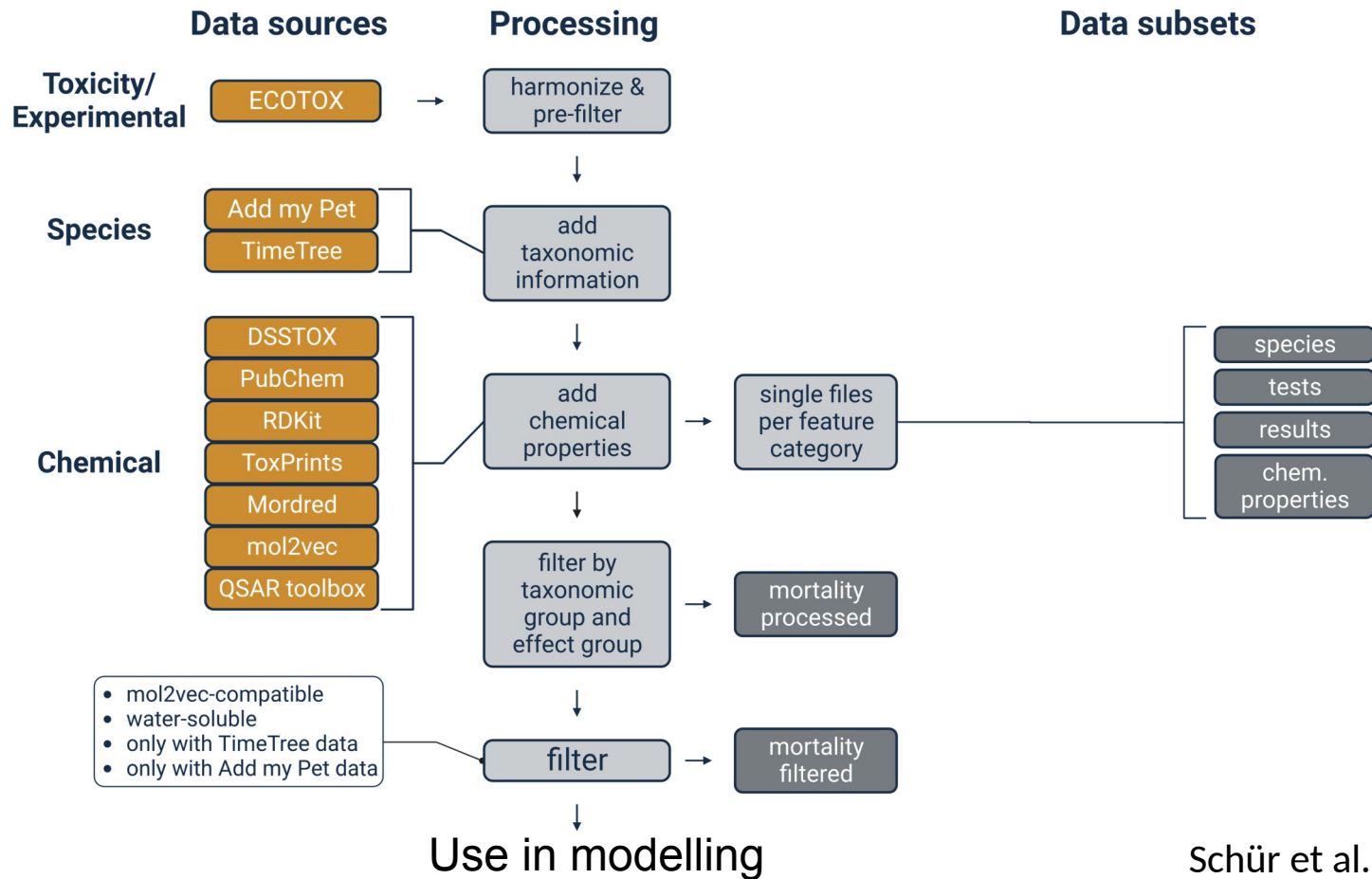
https://www.bio.vu.nl/thb/deb/deblab/add_my_pet/species_list.html



Open Traits Network

<https://opentraits.org/datasets.html>

Data pre-processing workflow



QSAR & RA tools



- QSARs
- Mixtures
- Hazard & Exposure
- TK
- USA EPA QSARs included
- ...

<https://arnotresearch.com/eas-e-suite/#>



- QSARs & RA
- Empirical data
- Contributions by US EPA, EFSA etc.
- ...

<https://qsartoolbox.org>



- QSARs & RA
- Empirical data
- Platform for many models
- ...

<https://www.vegahub.eu>

QSAR & RA tools

CompTox Chemicals Dashboard

Search 1,218,248 Chemicals

- Includes WebTEST
- QSAR & RA
- USA EPA
- ...

<https://comptox.epa.gov/>

OPERA

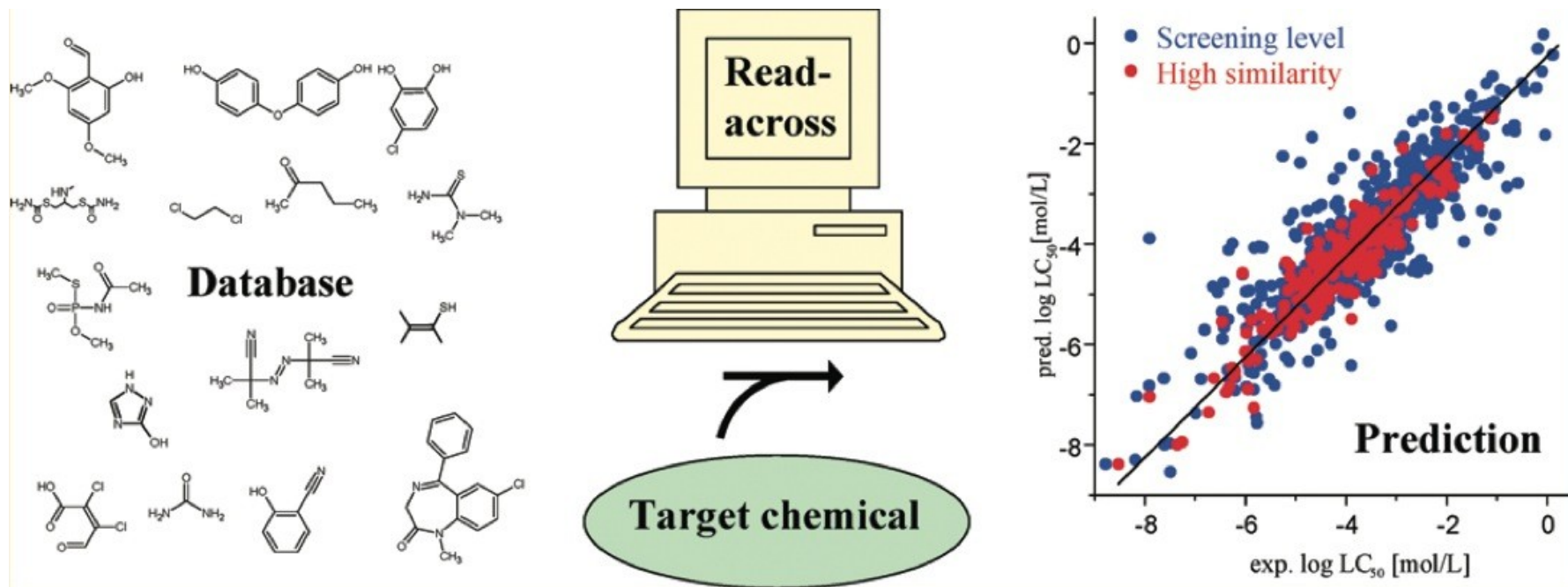
OPERA is a free and open-source/open-data suite of QSAR models providing predictions on physicochemical properties, environmental fate and toxicity endpoints as well as additional information including applicability domain and accuracy assessment. All models were built on curated data and standardized QSAR-ready chemical structures. OPERA is available in command line and user-friendly graphical interface for Windows and Linux operating systems. It can be installed as a standalone desktop application or embedded in a different tool/workflow.

- Suite of open source QSARs
- Originally developed by US EPA and NIEHS
- ...

<https://github.com/kmansouri/OPERA>

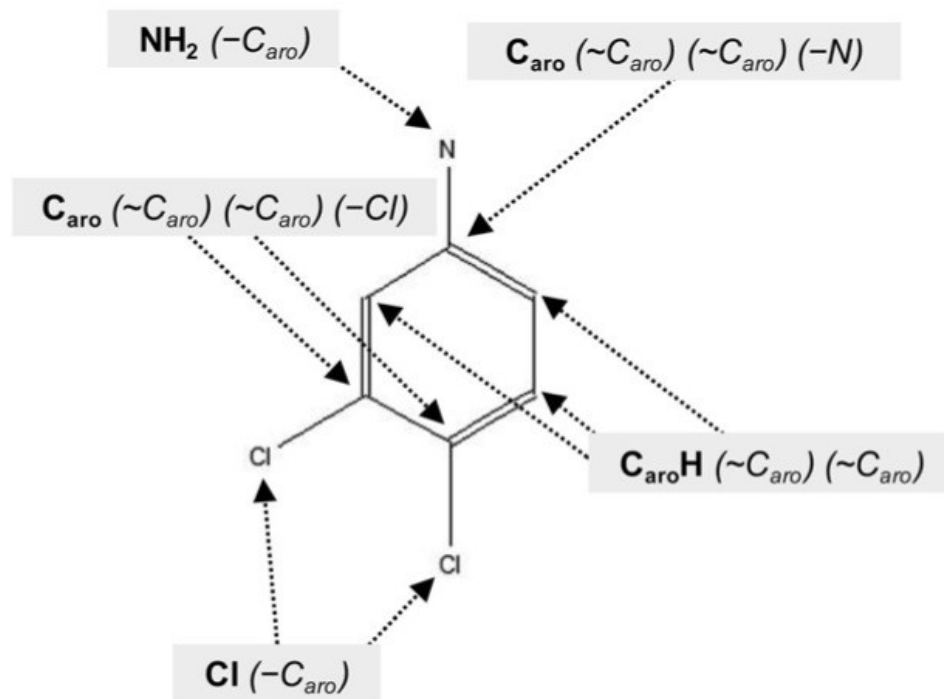
Quantitative Read-Across: Concept

Estimates toxicity based on similarity with compounds of known toxicity



Quantitative Read-Across: Concept

Similarity based on atomic-centred fragments (ACFs)



Calculation of Tanimoto similarity

1) Convert molecules to bitstrings

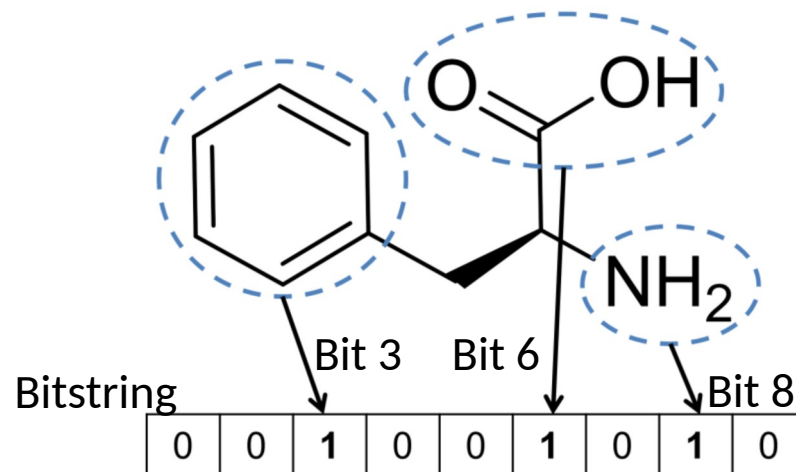
Example:

Molecule A: 0 0 1 0 0 1 0 1 0

Molecule B: 1 1 0 1 0 1 1 0 0

2) Compute similarity measure

		Chemical 1	
		Bit _i present	Bit _i absent
Chemical 2	Bit _i present	a	b
	Bit _i absent	c	d



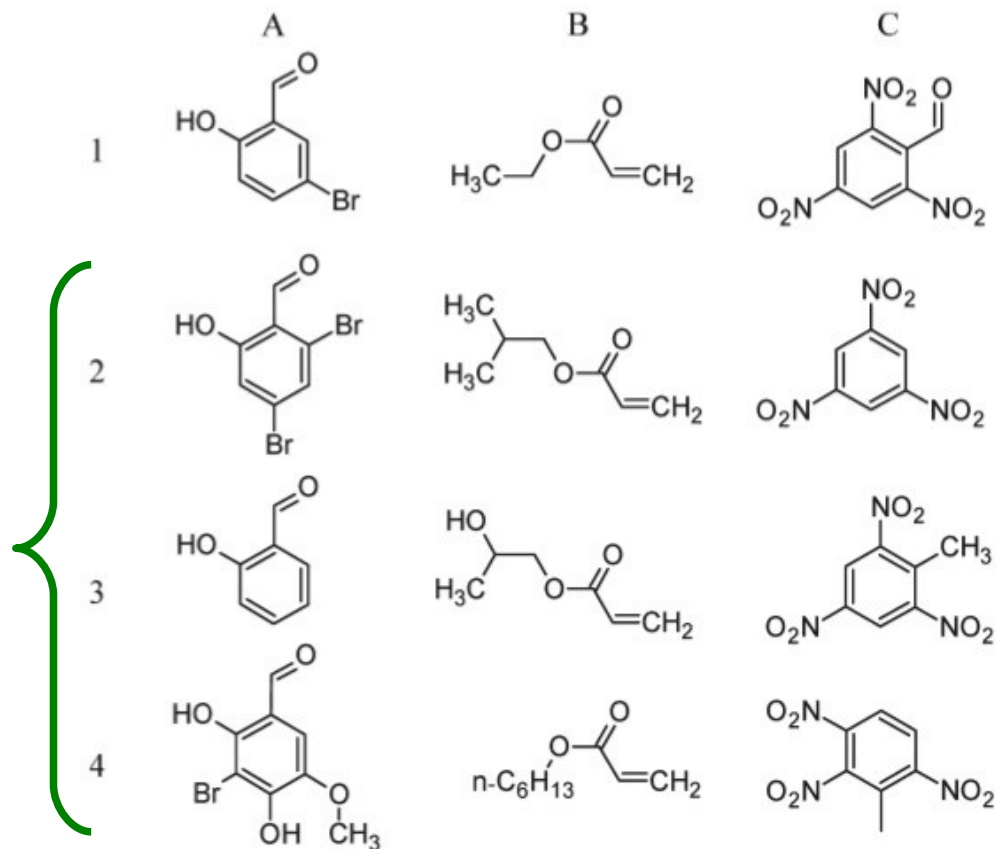
$$S_{\text{Tanimoto}} = \frac{a}{a+b+c}$$

Quantitative Read-Across: Example

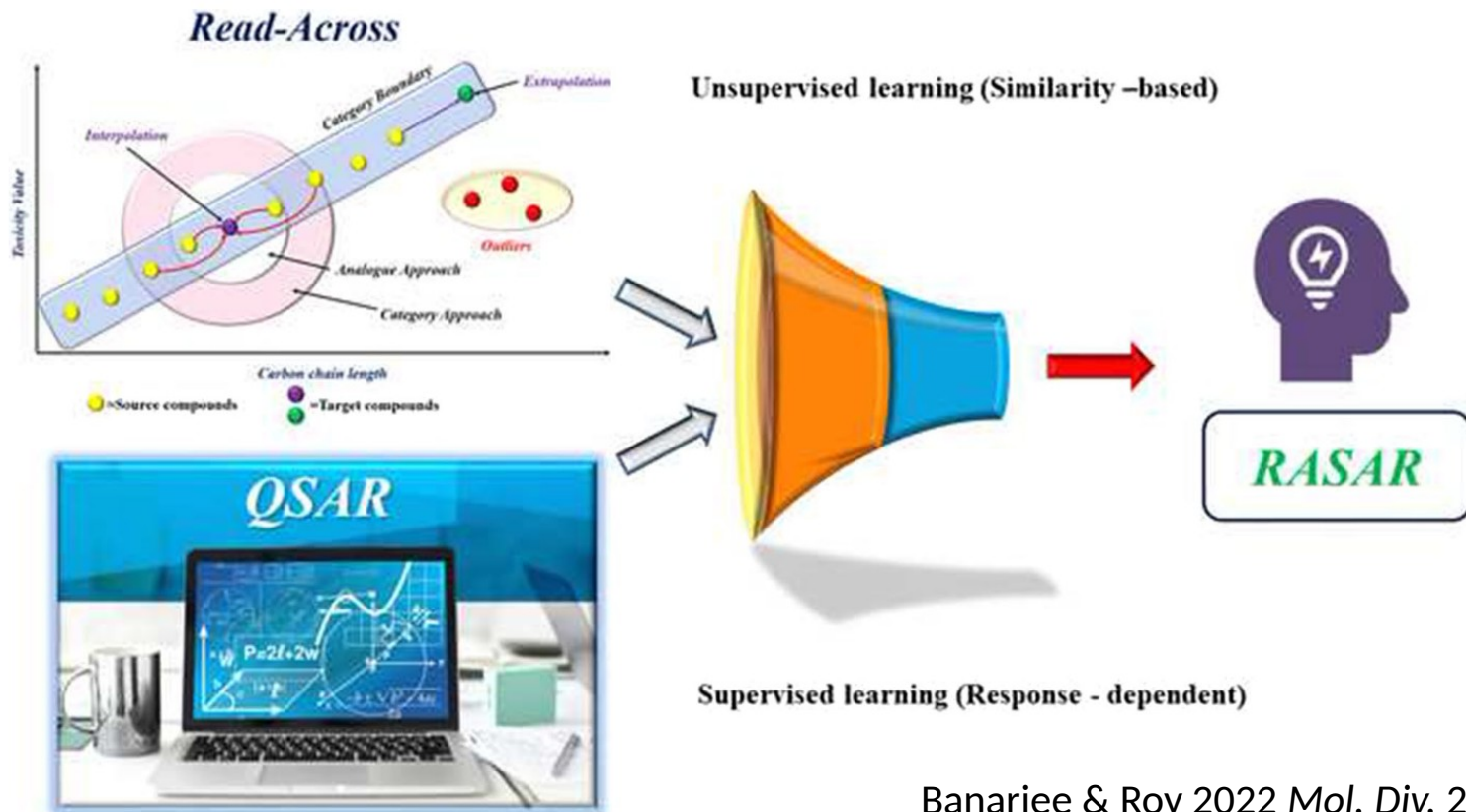
Prediction of properties (e.g. physico-chemical, toxicity) weighted by similarity

Target compounds

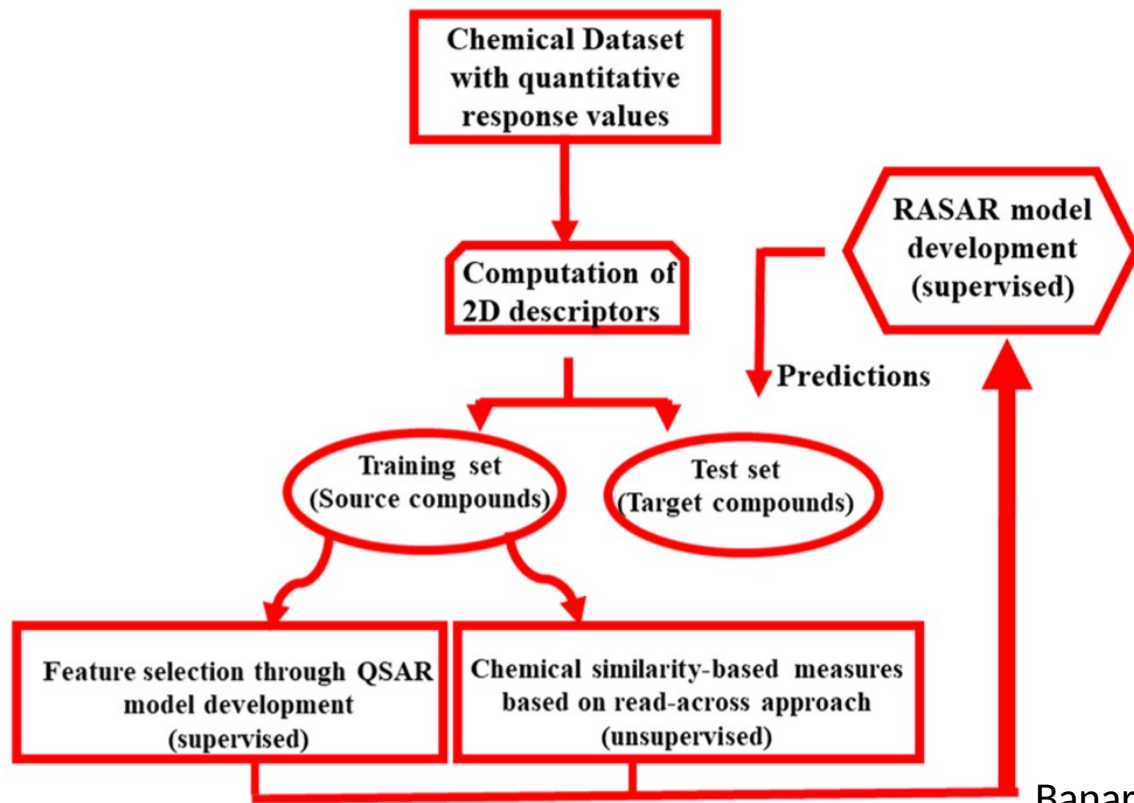
Compounds with known toxicity in decreasing similarity



qRASAR: Quantitative Read-Across structure-activity relationships



qRASAR: Quantitative Read-Across structure-activity relationships



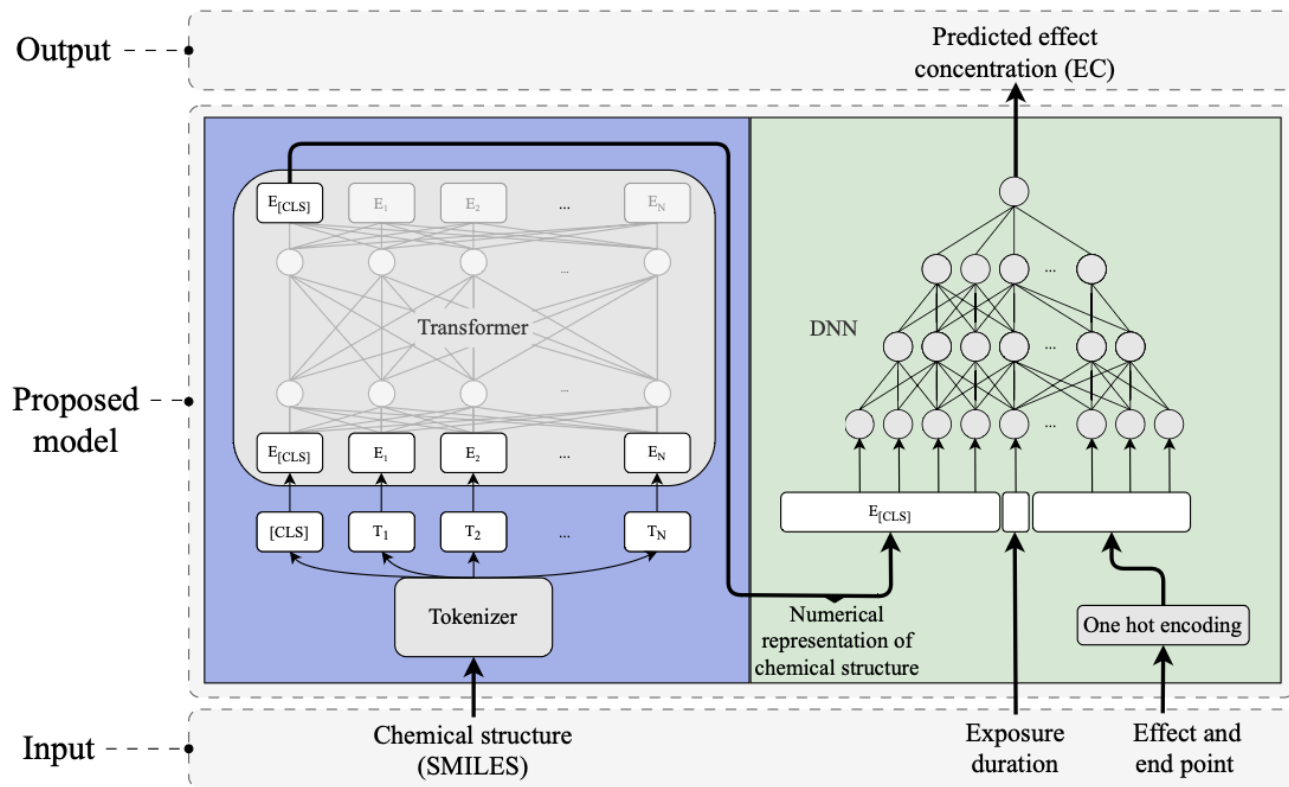
DTC Lab Tools

- Prediction tools
- RA
- RASAR
- ...

<https://sites.google.com/jadavpuruniversity.in/dtc-lab-software/home>

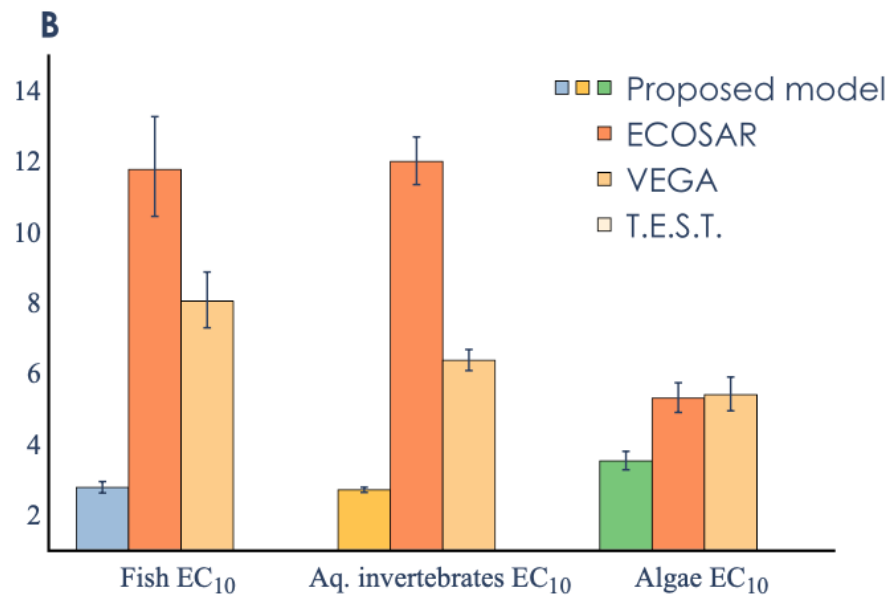
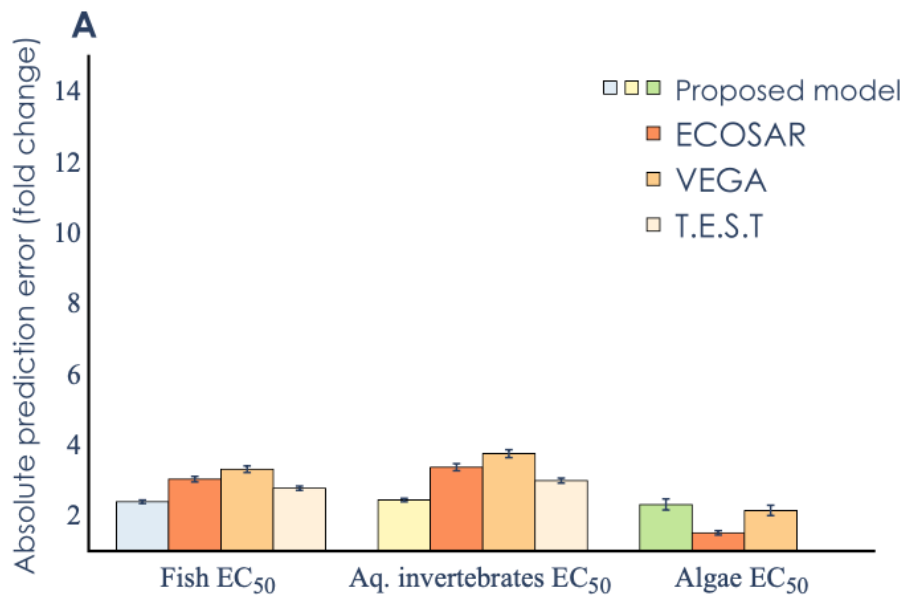
Case study: TRIDENT

TRIDENT: Based on LLM (BERT) and deep learning



Case study: TRIDENT

Generally better performance, in particular for EC₁₀



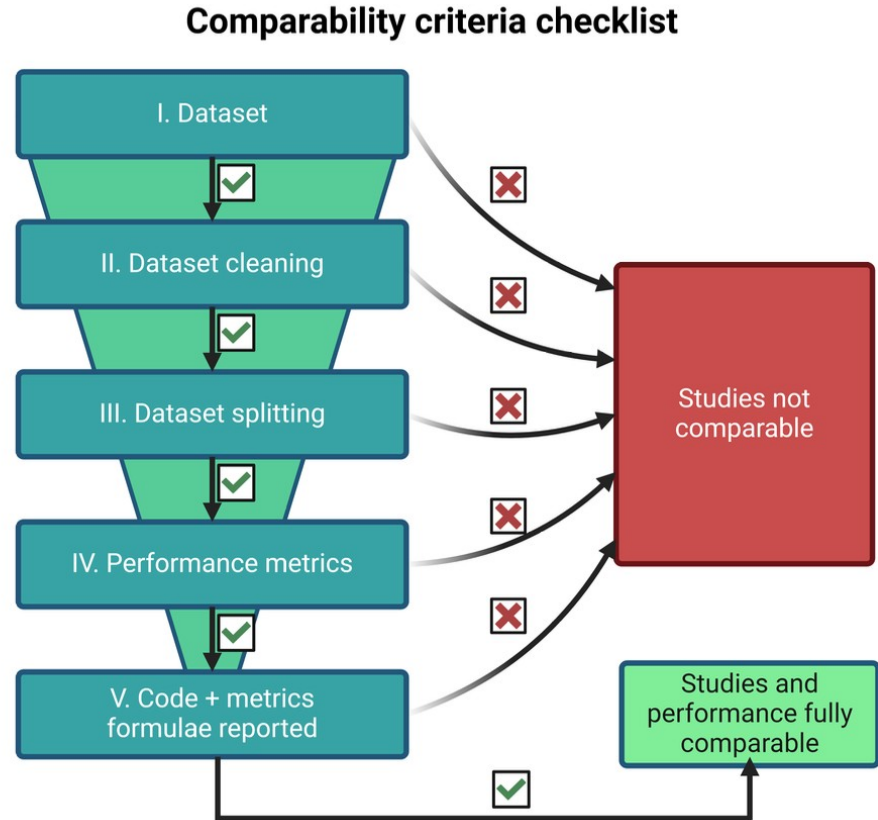
TRIDENT
Predicting chemical ecotoxicity using AI

<https://ecocait.streamlit.app/>

Gustavsson et al. 2024 *Sci. Adv.* 10: eadk6669

Validation of prediction models

What is the best model?
Unfortunately,
comparability is low

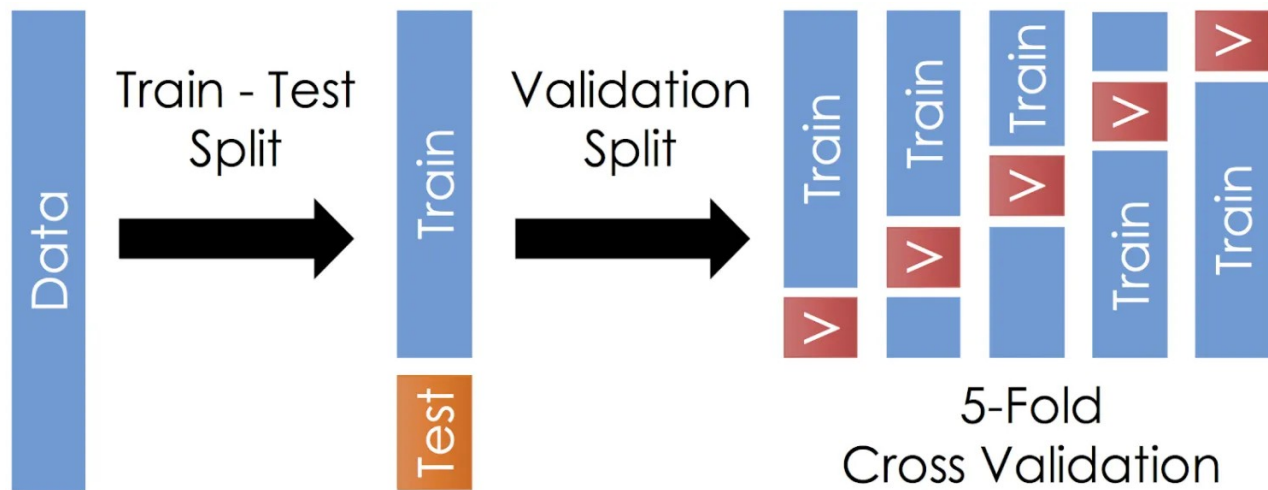


Example: Splitting methods

Validation typically done with n -fold cross-validation

Typical splits

- 1) Random
- 2) By compound
- 3) By scaffold



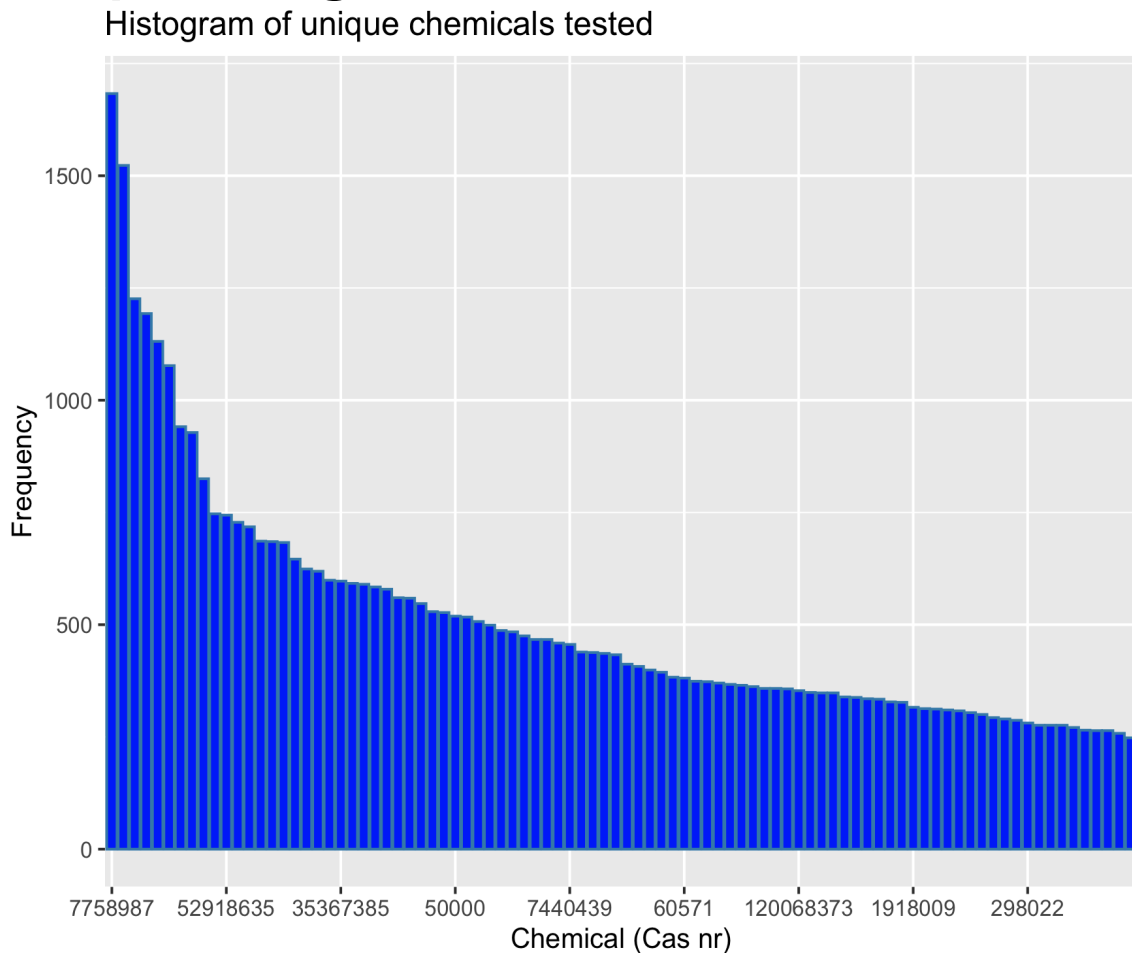
<https://medium.com/@hi.martinez/train-test-split-cross-validation-you-b87f662445e1>

Example: Splitting methods

Typical splits

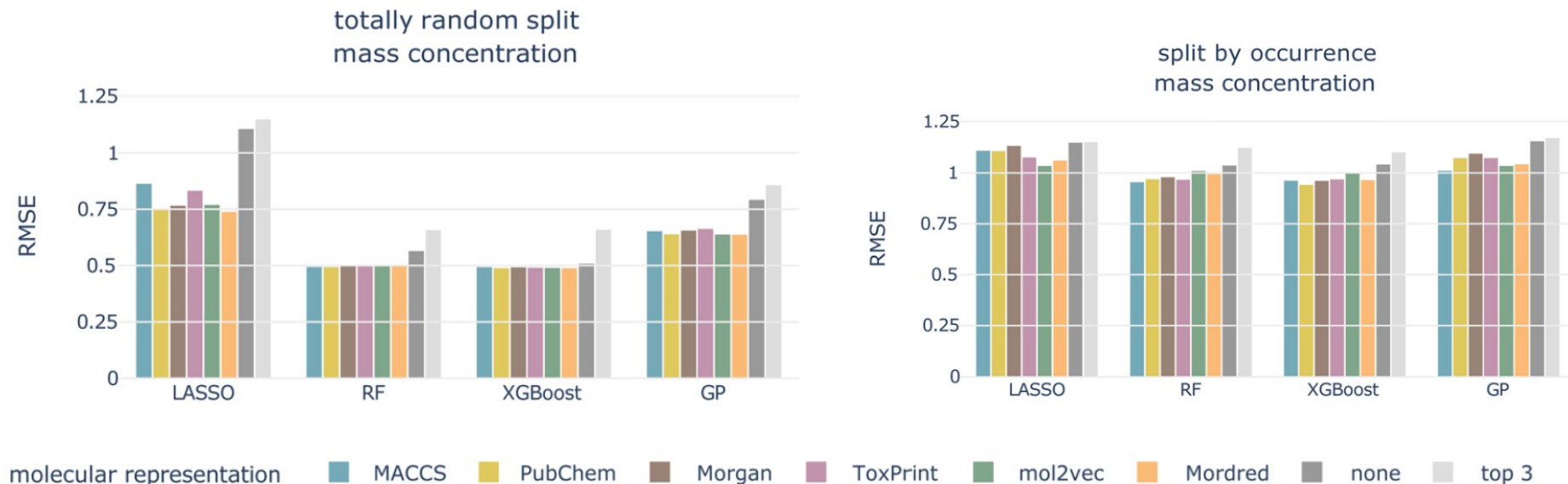
- 1) Random
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Random not recommended because high probability of same chemicals in test and training set

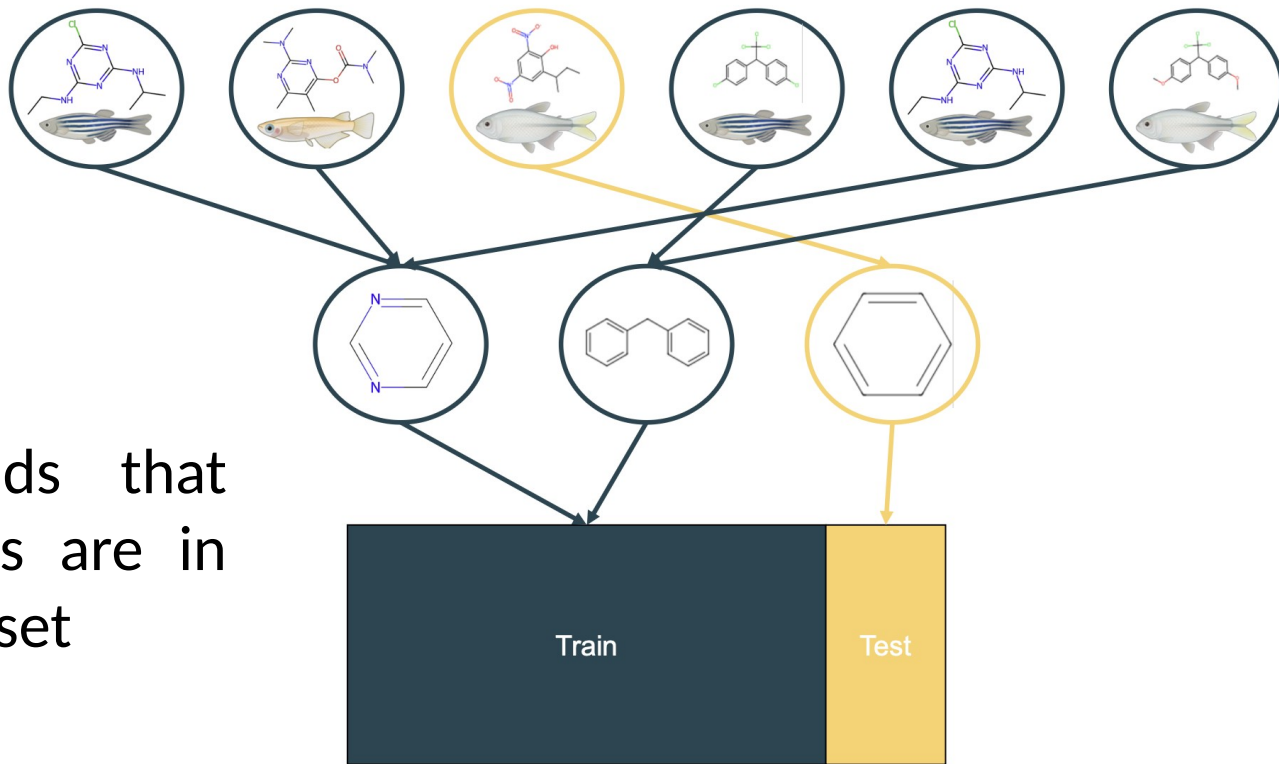


Example: Splitting methods

Much better performance for random split → too optimistic



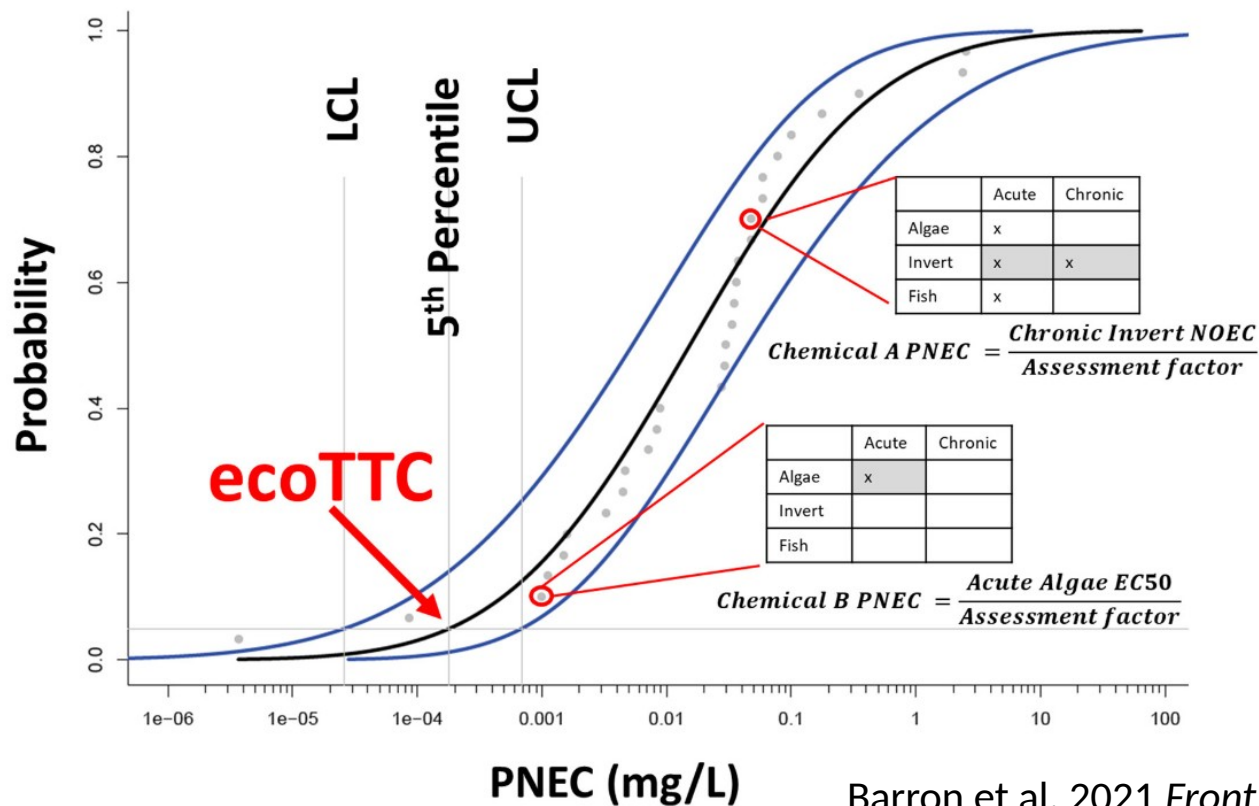
Example: Splitting methods



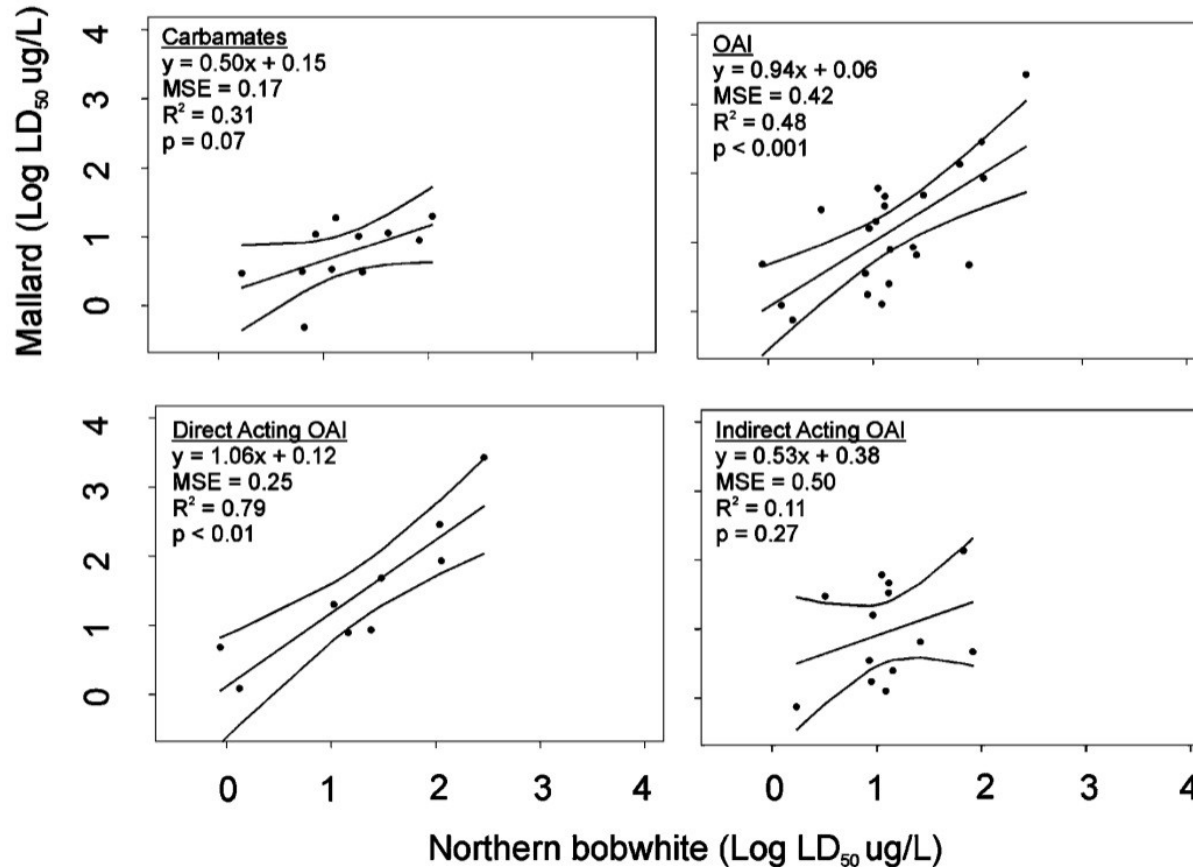
Scaffolding avoids that similar structures are in training and test set

EcoTCC: Ecol. Thresholds of Toxicological Concern

Screening tool for chemicals w/o toxicity data or outside QSAR domain



Interspecies Correlation Estimate (ICE)



- ICE for aquatic and terrestrial species
- SSDs
- ...

<https://sites.google.com/jadavpuruniversity.in/dtc-lab-software/home>

Acute-to-Chronic ratio (ACR)

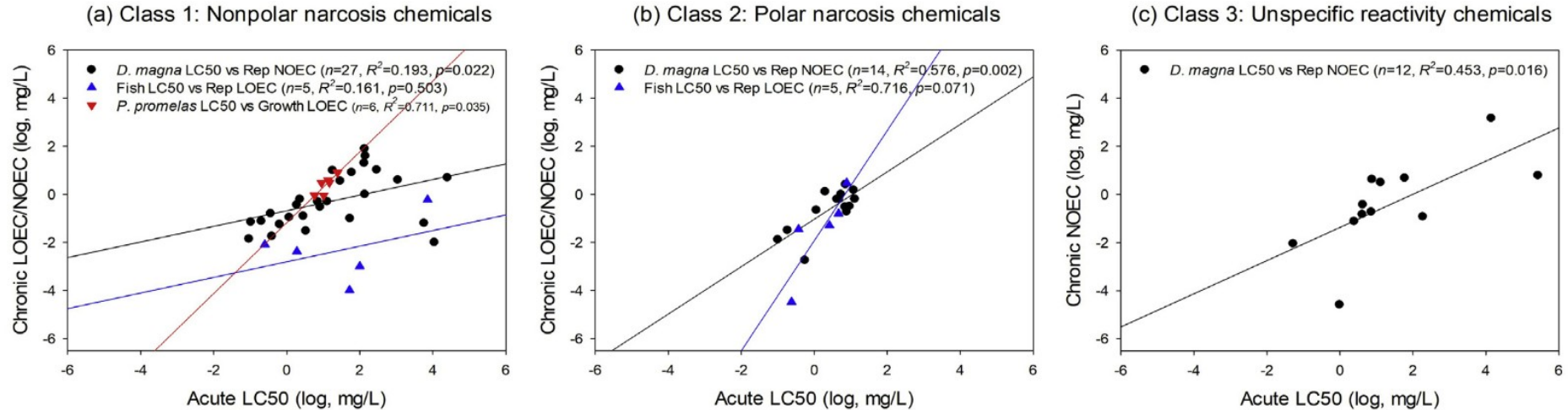
$$ACR = \frac{ECx(acute)}{ECx(chronic)}$$

Table 1. Overview of results from acute to chronic ratio (ACR) evaluation for each trophic level

Parameter	Green algae	<i>Daphnia magna</i>	Fish
No. of values	104	102	39
No. of substances	102	102	32
50th percentile (median)	5.4	7.0	10.5
90th percentile	33.3	41.5	198.2
Minimum value	1.06	0.6	1.3
Maximum value	4,400	2,222	4,250

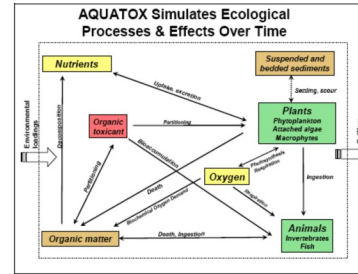
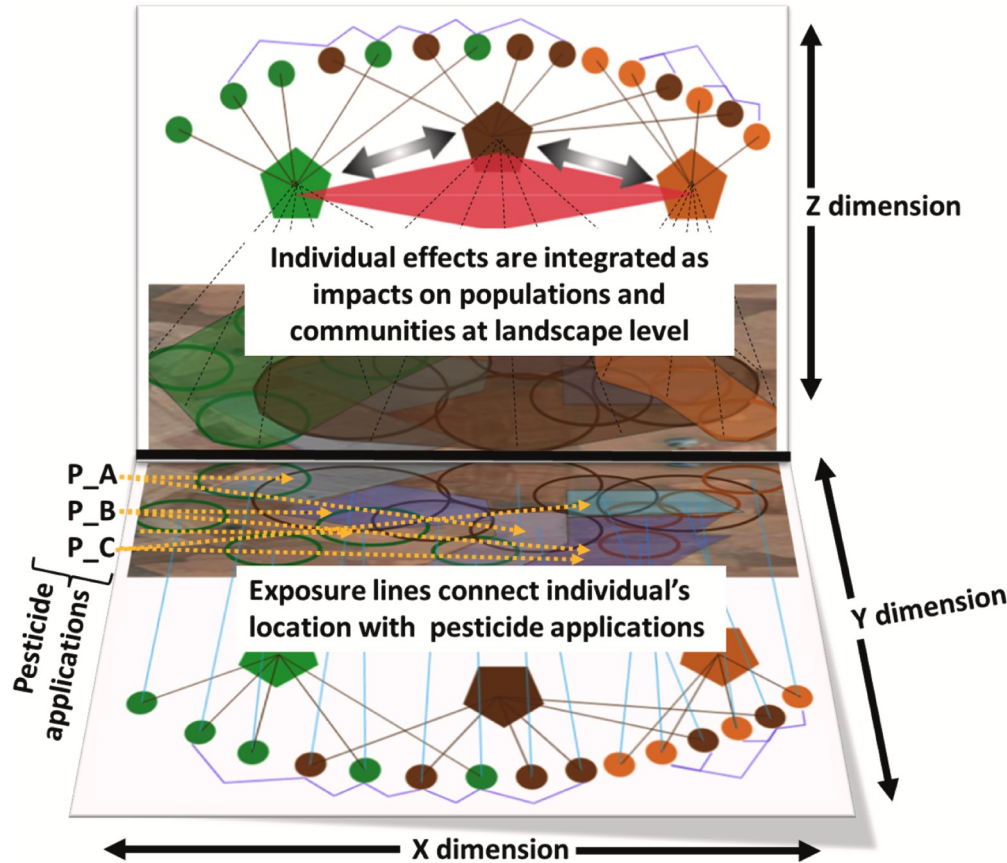
Acute-to-Chronic ratio (ACR)

ACRs dependent on MOA, organism group and test data properties



Eco(toxico)logical models

Wide range of general and specific eco(toxicological) models available



<https://www.epa.gov/hydrowq/aquatox>



<https://projects.au.dk/almass>

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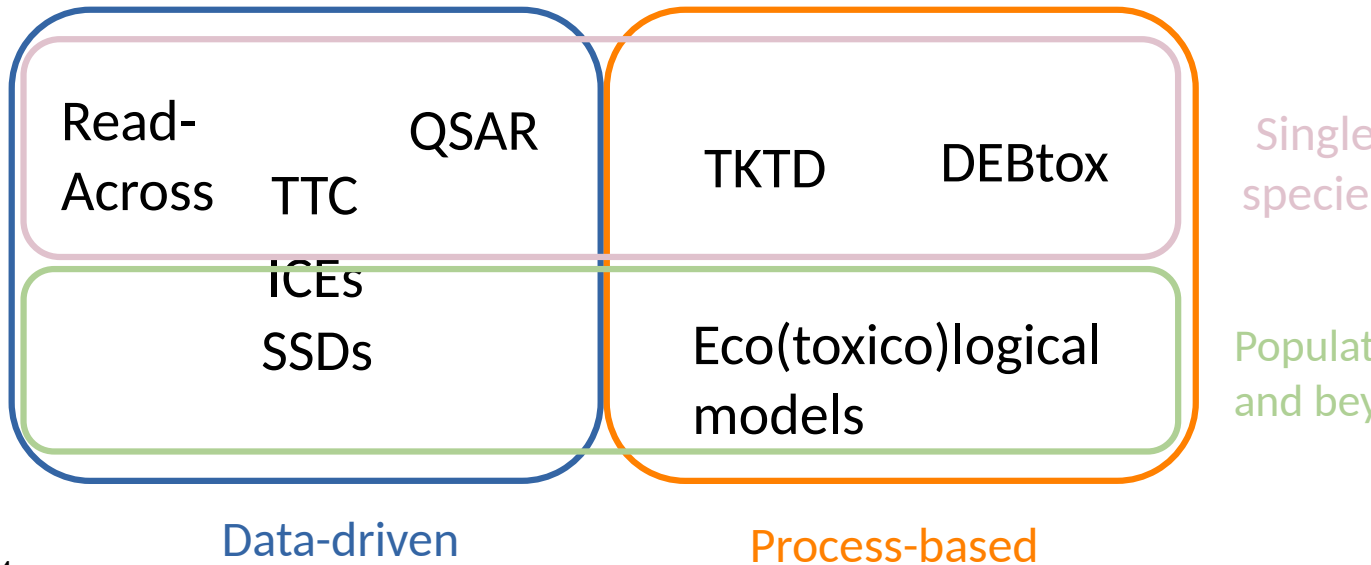
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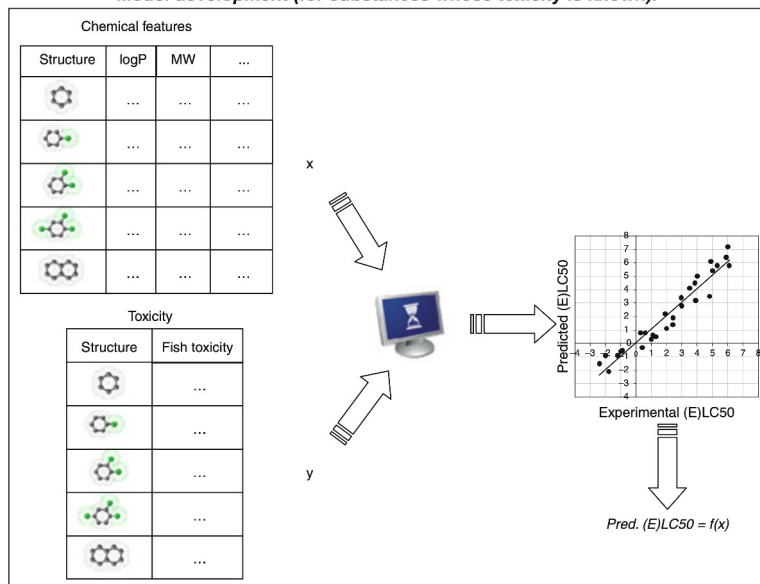
a , b , c and d = model parameters/ functions for variables

Congeneric chemicals share the same core chemical structure, but have different substituents. They typically have similar physicochemical and biological properties.

Examples: PCBs share same biphenyl structure, PAHs are characterised by multiple fused aromatic rings and PFAS share a perfluorinated carbon chain.

General QSAR development

Model development (for substances whose toxicity is known).

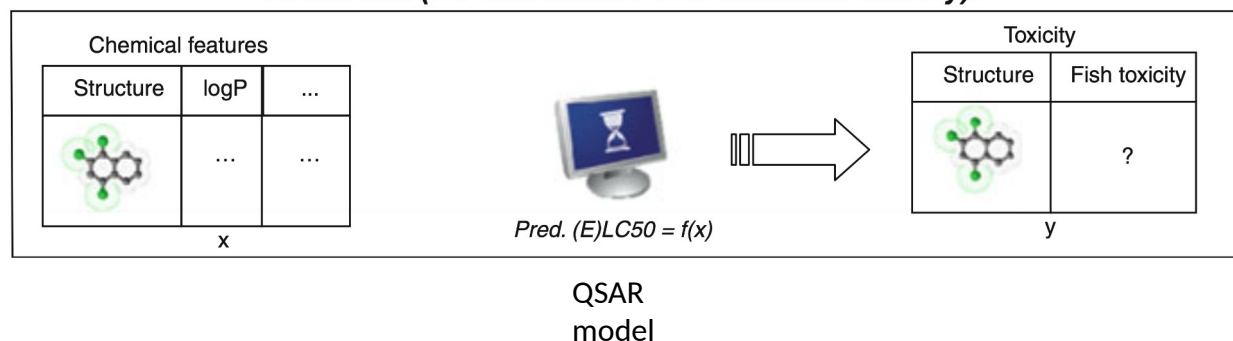


Lomborda et al. 2013

Lombardo A., Schifanella O., Roncaglioni A., Benfenati E. 2010: Quantitative Structure-Activity Relationship (QSAR) in Ecotoxicology. In: Férard J., Blaise C. (Ed.) Encyclopedia of Aquatic Ecotoxicology: SpringerReference (www.springerreference.com). Springer-Verlag Berlin Heidelberg, 2013. DOI: 10.1007/SpringerReference_380001

General QSAR development

Model use (for substances with unknown toxicity).



Prediction can be based on classical statistical (regression) or machine learning model

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Input data for *in silico* approaches

Typical input formats (chemicals)

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IUPAC name	1,3,7-trimethylpurine-2,6-dione

Canonical SMILES (Simplified Molecular Input Line Entry System), but not standard SMILES, and the InChI (International Chemical Identifier (IUPAC-based)) are unique and capture stereochemistry. The InChIkey is a hashed version of InChI and is more suitable for indexing and searching, but cannot be reversed to structure.

Input data for *in silico* approaches

Typical chemical input data: Chemical descriptors

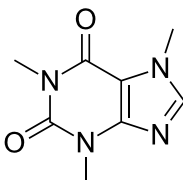
1-D

Scalar
physicochemical

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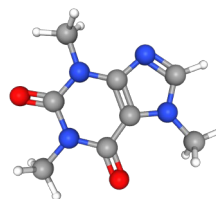
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- No. of atomic fragments (e.g. halogens, rings)
- Connectivity
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3-D

Spatial/electron distribution



- Steric fields
- Electrostatic maps
- ...

Many sources provide chemical descriptors (e.g. Mordred, Morgan, mol2vec), for an overview see: Schür, C., Gasser, L., Perez-Cruz, F., Schirmer, K., & Baity-Jesi, M. (2023). A benchmark dataset for machine learning in ecotoxicology. *Scientific Data*, 10(1), 718. <https://doi.org/10.1038/s41597-023-02612-2>

Input data for *in silico* approaches

Typical chemical input data: Biological descriptors

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- volume-specific somatic maintenance
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Add-my-Pet



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Phylogeny

e.g. **ape**, **taxize** R packages

Traits

- Feeding type
- Life span
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- ...

Biogeography

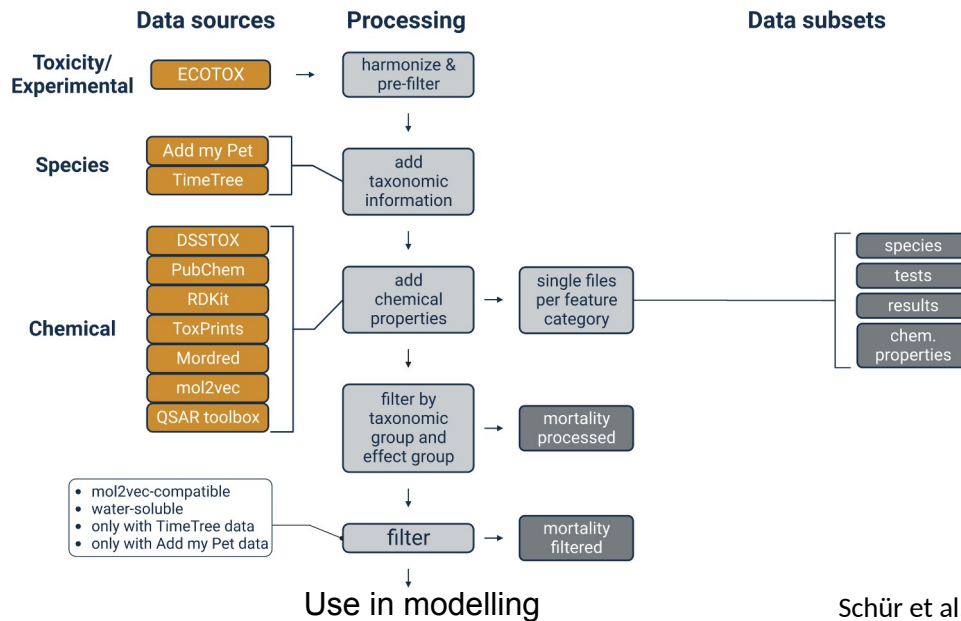
- Ecozone (e.g. geology, altitude)
- Climate
- ...



Open Traits Network

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Data pre-processing workflow



QSAR & RA tools



- QSARs
- Mixtures
- Hazard & Exposure
- TK
- USA EPA QSARs included

• ...

<https://arnotresearch.com/eas-e-suite/#>

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- QSARs & RA
- Empirical data
- Contributions by US EPA, EFSA etc.
- ...

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- QSARs & RA
- Empirical data
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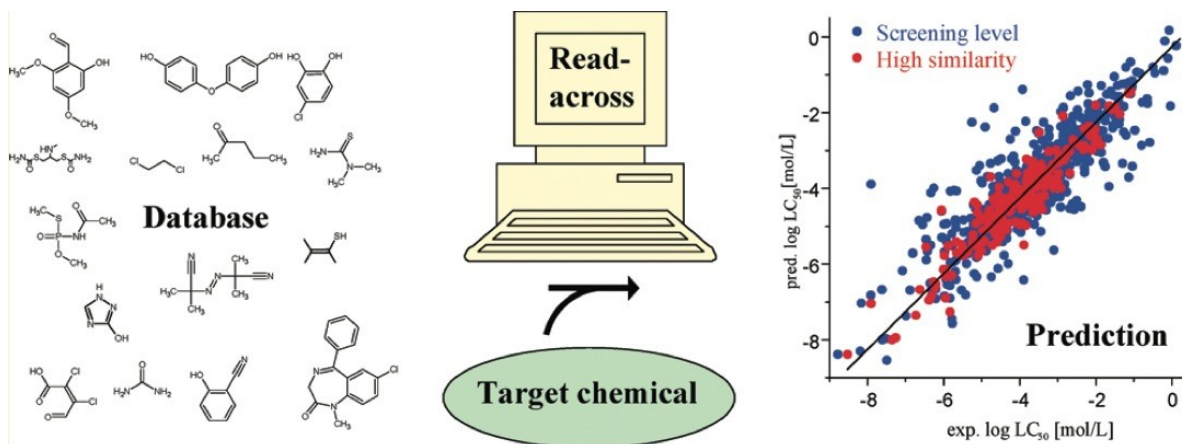
- Suite of open source QSARs
- Originally developed by US EPA and NIEHS
- ...

<https://github.com/kmansouri/OPERA>

You can download a local version of the Toxicity Estimation Software Tool (TEST) under:
<https://www.epa.gov/comptox-tools/toxicity-estimation-software-tool-test>

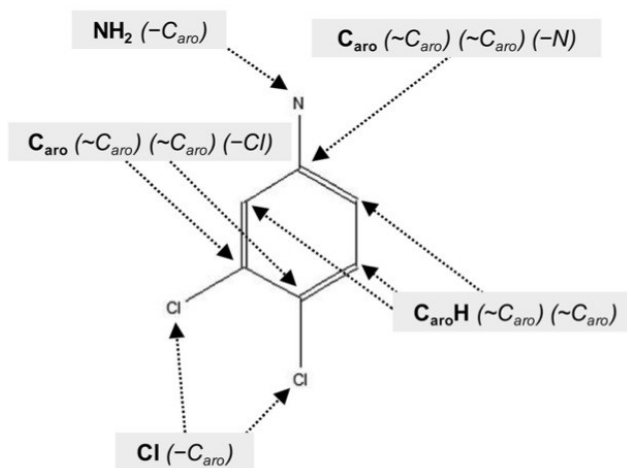
Quantitative Read-Across: Concept

Estimates toxicity based on similarity with compounds of known toxicity



Quantitative Read-Across: Concept

Similarity based on atomic-centred fragments (ACFs)



The chemical is decomposed into fragments, which are used to compute the similarity between compounds.

Calculation of Tanimoto similarity

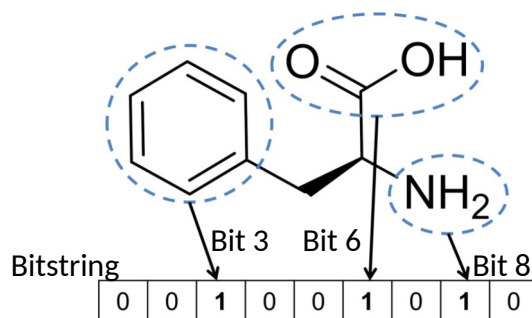
1) Convert molecules to bitstrings

Example:

Molecule A: 0 0 1 0 0 1 0 1 0

Molecule B: 1 1 0 1 0 1 1 0 0

2) Compute similarity measure



$$S_{\text{Tanimoto}} = \frac{a}{a+b+c}$$

		Chemical 1	
		Bit _i present	Bit _i absent
Chemical 2	Bit _i present	a	b
	Bit _i absent	c	d

18

Bajusz et al. 2017

The Tanimoto similarity is identical to the Jaccard similarity in ecology.

$a = 1, b = 2, c = 4$

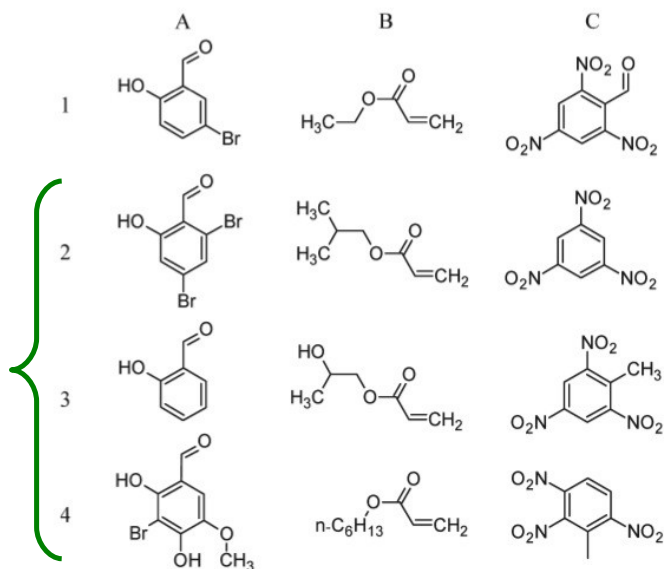
Bajusz, D., Rácz, A., & Héberger, K. (2017). Chemical Data Formats, Fingerprints, and Other Molecular Descriptions for Database Analysis and Searching. In *Comprehensive Medicinal Chemistry III* (pp. 329–378). Elsevier. <https://doi.org/10.1016/B978-0-12-409547-2.12345-5>

Quantitative Read-Across: Example

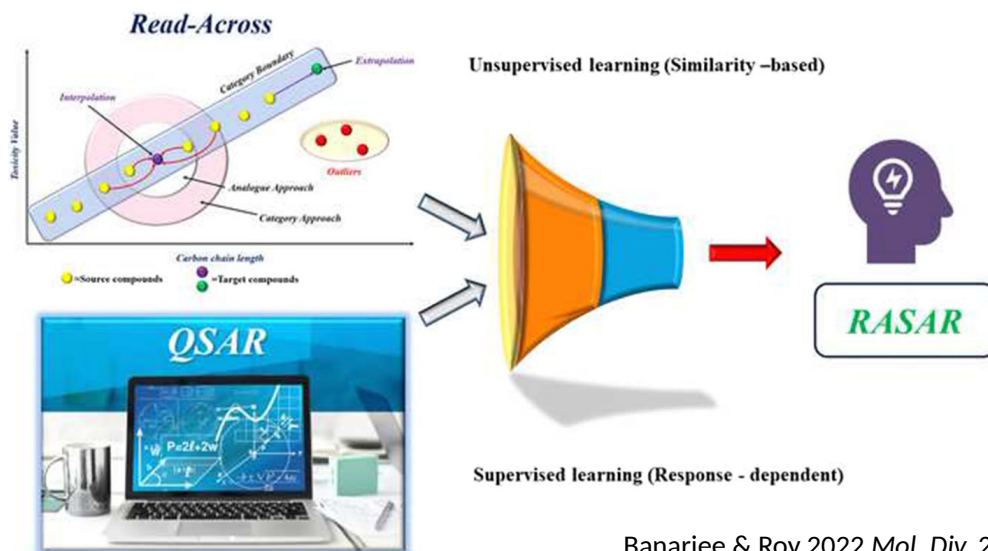
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Target compounds

Compounds with known toxicity in decreasing similarity

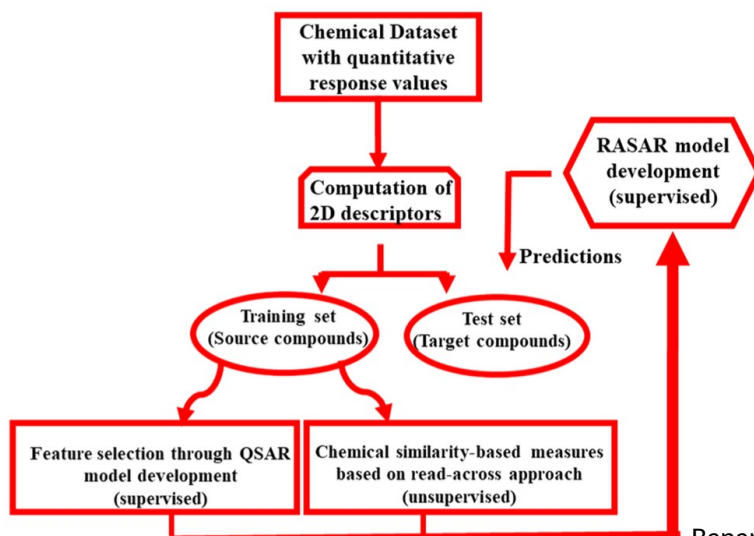


qRASAR: Quantitative Read-Across structure-activity relationships



Banarjee & Roy 2022 Mol. Div. 26: 2847

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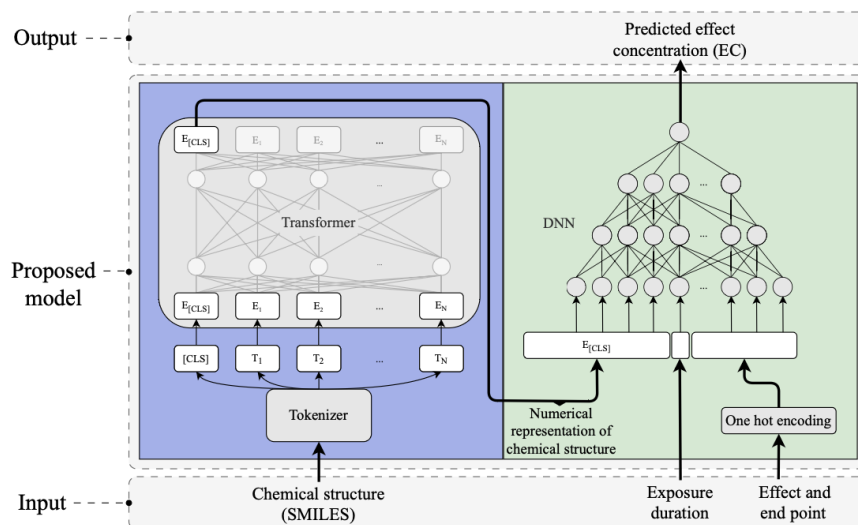
DTC Lab Tools

- Prediction tools
- RA
- RASAR
- ...

<https://sites.google.com/jadavpuruniversity.in/dtc-lab-software/home>

Case study: TRIDENT

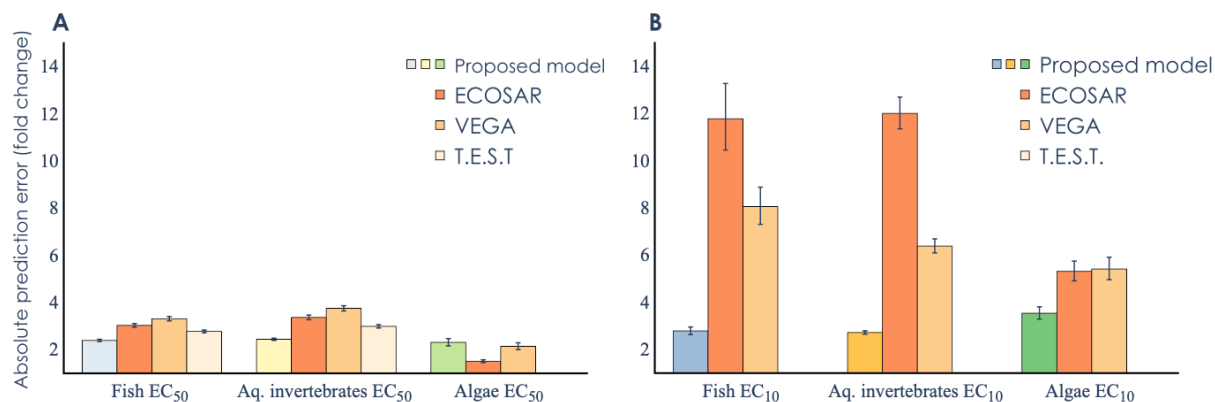
TRIDENT: Based on LLM (BERT) and deep learning



The background of the language processing is explained here along with the computer code:
<https://deepchem.io/tutorials/transfer-learning-with-chemberta-transformers/>

Case study: TRIDENT

Generally better performance, in particular for EC₁₀



TRIDENT
Predicting chemical ecotoxicity using AI

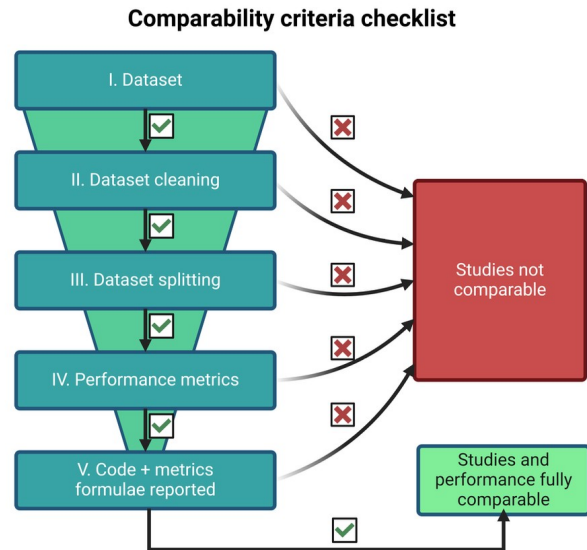
<https://ecocait.streamlit.app/>

For another application see the following study that incorporated a wider range of biological information:

Zubrod, J. P., Galic, N., Vaugeois, M., & Dreier, D. A. (2024). Bio-QSARs 2.0: Unlocking a new level of predictive power for machine learning-based ecotoxicity predictions by exploiting chemical and biological information. *Environment International*, 186, 108607. <https://doi.org/10.1016/j.envint.2024.108607>

Validation of prediction models

What is the best model?
Unfortunately,
comparability is low



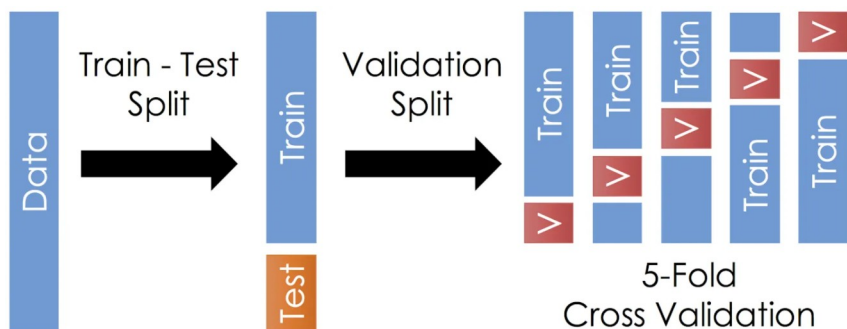
The preprint is available at:
<https://doi.org/10.1101/2025.03.04.641385>

Example: Splitting methods

Validation typically done with n -fold cross-validation

Typical splits

- 1) Random
- 2) By compound
- 3) By scaffold



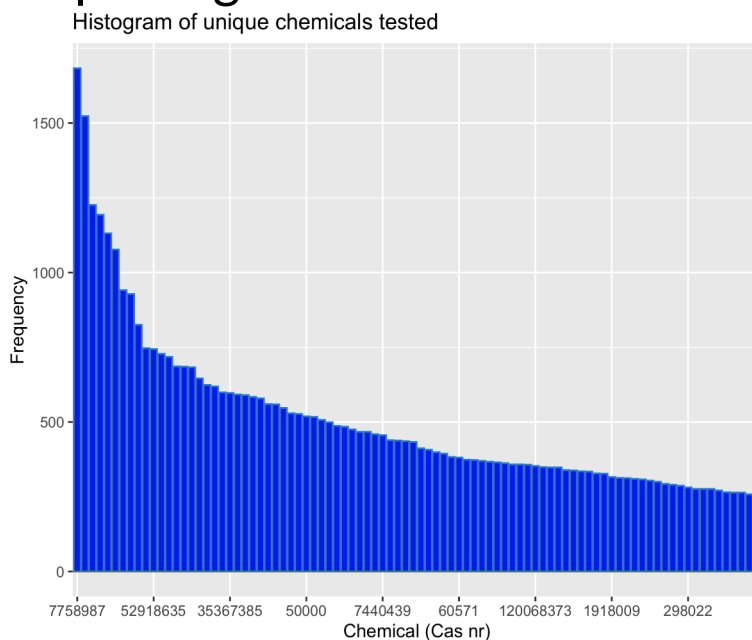
<https://medium.com/@hi.martinez/train-test-split-cross-validation-you-b87f662445e1>

Example: Splitting methods

Typical splits

- 1) Random
- 2) By compound
- 3) By scaffold

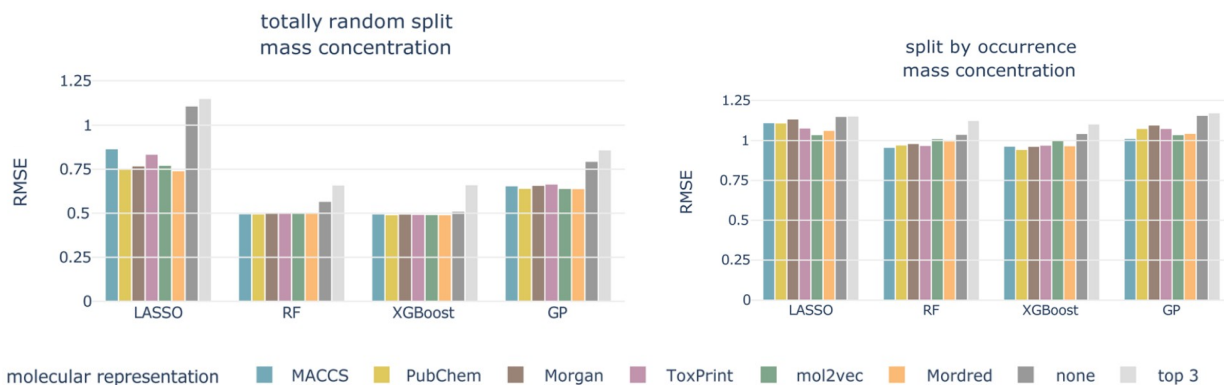
Random not recommended because high probability of same chemicals in test and training set



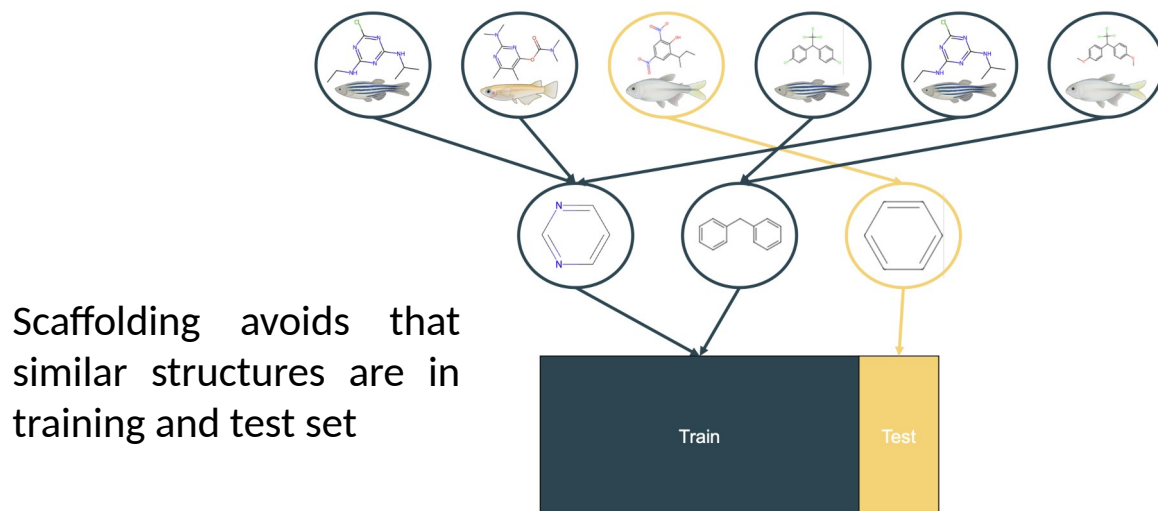
Own calculations of the number of tests per unique chemical in the EPA Ecotox dataset.

Example: Splitting methods

Much better performance for random split → too optimistic



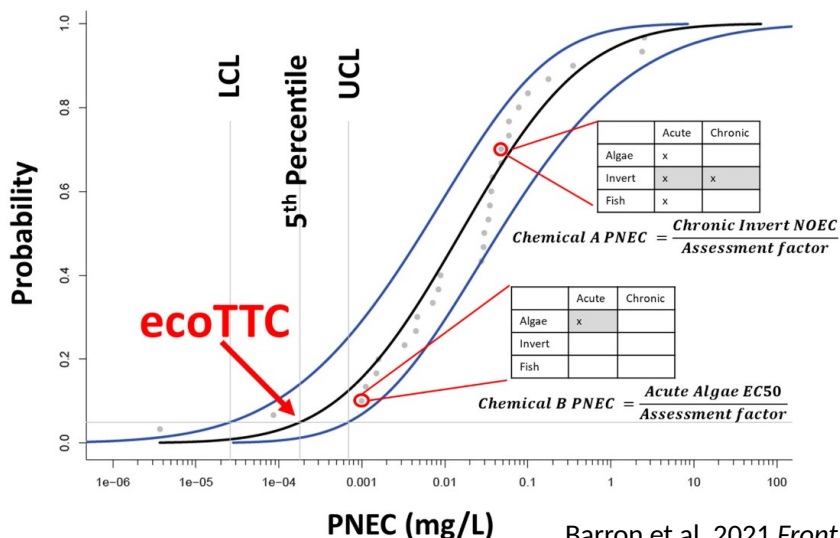
Example: Splitting methods



In case of prediction of compounds of a similar chemical class, scaffolding would neither be applicable (nor required).

EcoTCC: Ecol. Thresholds of Toxicological Concern

Screening tool for chemicals w/o toxicity data or outside QSAR domain

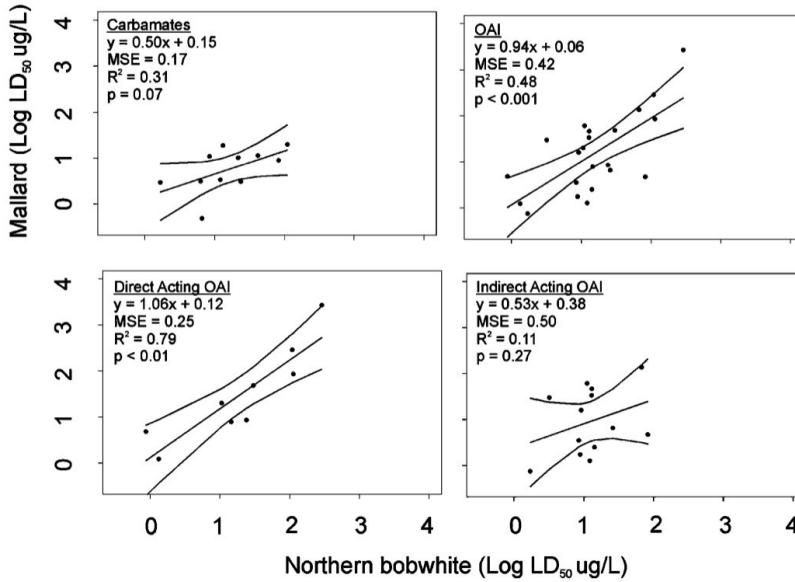


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Barron et al. 2021 *Front. Toxicol.* 3: 640183

A related tool is available in the EnviroTox database:
<https://envirottoxdatabase.org>

Interspecies Correlation Estimate (ICE)



- ICE for aquatic and terrestrial species
- SSDs
- ...

<https://sites.google.com/jadavpuruniversity.in/dtc-lab-software/home>

Acute-to-Chronic ratio (ACR)

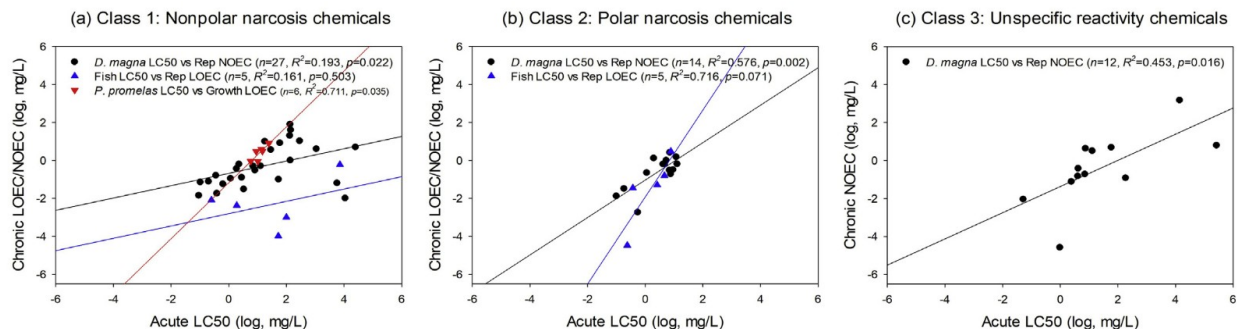
$$ACR = \frac{ECx(acute)}{ECx(chronic)}$$

Table 1. Overview of results from acute to chronic ratio (ACR) evaluation for each trophic level

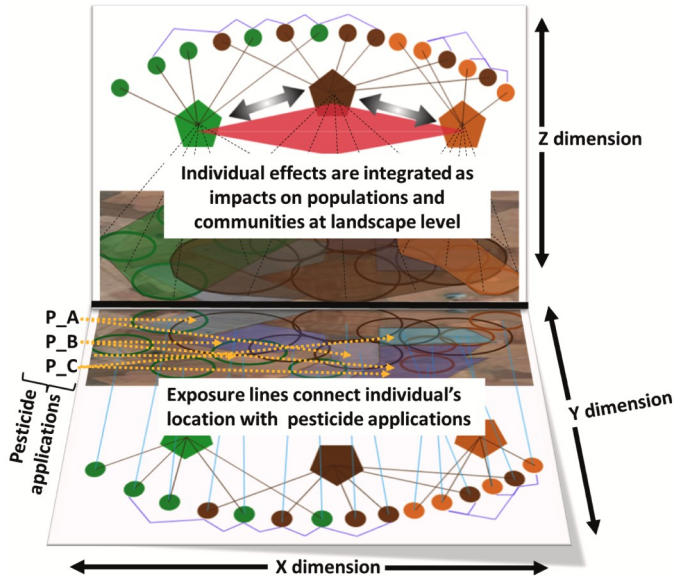
Parameter	Green algae	<i>Daphnia magna</i>	Fish
No. of values	104	102	39
No. of substances	102	102	32
50th percentile (median)	5.4	7.0	10.5
90th percentile	33.3	41.5	198.2
Minimum value	1.06	0.6	1.3
Maximum value	4,400	2,222	4,250

Acute-to-Chronic ratio (ACR)

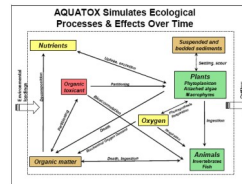
ACRs dependent on MOA, organism group and test data properties



Eco(toxico)logical models



Wide range of general and specific eco(toxicological) models available



<https://www.epa.gov/hydrowq/aquatox>



<https://projects.au.dk/almass>